

# Notes on Jermann and Quadrini (2012)

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**Motivation** 1. Financial sector was not well studied. 2. Studies did not simultaneously replicate debt and equity. This paper thinks that they should not be separated.

**Research Question** How are the dynamics of real and financial variables affected by "financial shocks"?

**Model** RBC Model + Financial friction

(a) Firm: Continuum of firms on  $[0, 1]$ . Gross revenue function  $F(z_t, k_t, n_t) = z_t k_t^\theta n_t^{1-\theta}$ .

Capital is chosen at time  $t - 1$  and labour is flexible at  $t$ . Capital evolves according to

$$k_{t+1} = (1 - \delta)k_t + i_t.$$

Firms use equity (payout  $d_t$ ) and debt ( $b_t$ ). Debt is preferred since its tax advantage.

Effective gross interest rate is  $R_t = 1 + r_t(1 - \tau)$  where  $\tau$  is tax.

Debt  $b_t$  is intertemporal, and loan  $\ell_t$  is intraperiod to finance working capital without interest. The loan is

$$\ell_t = w_t n_t + i_t + b_t + d_t - \frac{b_{t+1}}{R_t}.$$

Budget constraint:

$$b_t + d_t + w_t n_t + k_{t+1} = \frac{b_{t+1}}{R_t} + (1 - \delta)k_t + F(z_t, k_t, n_t).$$

Hence,  $\ell_t = F(z_t, k_t, n_t)$ .

- (b) Consider default. Firms has liability  $\ell_t + \frac{b_{t+1}}{R_t}$ , and liquidity  $\ell_t = F(z_t, k_t, n_t)$ . However, firm can divert<sup>1</sup> to pay debt. Then the creditors can only liquidate  $k_{t+1}$ . Given default, suppose that creditors have probability  $\xi_t$  of recovering the full value of  $k_{t+1}$  and otherwise zero. As a result, the expected recovery value is  $\xi_t k_{t+1}$ . The enforcement constraint is:

$$\xi_t \left( k_{t+1} - \frac{b_{t+1}}{1 + r_t} \right) \geq \ell_t. \quad (0.1)$$

Why we have  $1 + r_t$  instead of effective rate?  $b_t/(1 + r_t)$  is the market value of the debt at time  $t$  so it is subtracted to compute the residual collateral.  $b_{t+1}/R_t$ , however, is the effective borrowed amount. It is less than the market value.

So what is the interpretation of this (0.1)? It sets a credit limit: loan will not be available if this condition is not met, or, equivalently, the intratemporal loan amount will not exceed the credit limit. The stochastic  $\xi_t$  captures the financial market uncertainty. Comparably in my model, it is the second-order productivity shock that is exogenous and controlling the tightness of the credit constraint.

Rewriting the enforcement constraint using the budget constraint, we have

$$\frac{\xi_t}{1 - \xi_t} [(1 - \delta)k_t - b_t - w_t n_t - d_t] \geq F(z_t, k_t, n_t).$$

Therefore, if there is a negative financial shock, that is,  $\xi_t$  decreases, then conditional on that the production plan does not change, the firm will have to increase the equity, that is, cut the payout  $d_t$ . However, if  $d_t$  cannot be cut more, then labour will be reduced.

- (c) Firm's recursive problem is

$$V(\mathbf{s}; k, b) = \max_{d, n, k', b'} \{d + E m' V(\mathbf{s}'; k', b')\}$$

subject to

$$(1 - \delta)k + F(z, k, n) - wn + \frac{b'}{R} = b\varphi(d) + k'$$

$$\xi \left( k' - \frac{b'}{1 + r} \right) \geq F(z, k, n),$$

where  $\varphi(d) = d + \kappa(d - \bar{d})$  is equity payout cost.

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<sup>1</sup>Firms can make the revenue unavailable to creditors. This helps create limited enforcement.

- (d) Household: Homogenous agents maximizing lifetime utility  $\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(c_t, n_t)$ , subject to the budget constraint

$$w_t n_t + b_t + s_t(d_t + p_t) = \frac{b_{t+1}}{1 + r_t} + s_{t+1}p_t + c_t + T_t,$$

where  $s_t$  is the equity share and  $T_t$  is a lump-sum tax that finances the firm debt tax benefit.

## References

**Jermann, Urban and Vincenzo Quadrini**, “Macroeconomic Effects of Financial Shocks,” *American Economic Review*, February 2012, 102 (1), 238–271.